Phase Interpolators

From I/Q First Principles to SerDes & Beam Steering

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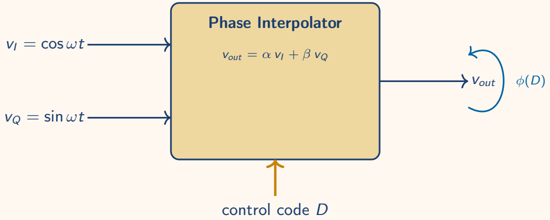
First Principles

What is a Phase Interpolator?

A circuit that produces a clock at any phase between two reference phases by taking a weighted sum.

- **Inputs:** two reference clocks, same frequency, known phase offset (typically 90° apart — I and Q)
- **Control:** digital code $\rightarrow$ analog weights (α, β)
- **Output:** one clock with phase $\phi \in [\phi_I, \phi_Q]$, same frequency

No frequency synthesis, no PLL — *just controlled mixing of two phases*.



Black box: 2 phases in, 1 phase out, code controls where.

First Principles — I/Q Weighted Sum

Given two orthogonal clocks at ω:

$$v_I(t) = A\cos\omega t, \qquad v_Q(t) = A\sin\omega t.$$

Take a weighted sum with real weights α, β:

$$v_{out}(t) = \alpha\, v_I(t) + \beta\, v_Q(t) = A\big[\alpha\cos\omega t + \beta\sin\omega t\big].$$

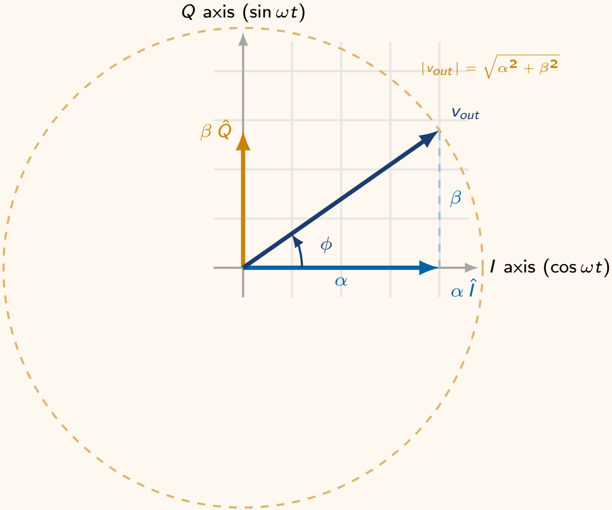
Apply the harmonic-addition identity $a\cos\theta + b\sin\theta = R\cos(\theta - \phi)$ with $R = \sqrt{a^2 + b^2}$, $\tan\phi = b/a$:

$$\boxed{v_{out}(t) = A\sqrt{\alpha^2 + \beta^2}\,\cos(\omega t - \phi)\,, \quad \phi = \arctan\frac{\beta}{\alpha}.}$$

Takeaway

The output *frequency* is unchanged. The output *phase* is set entirely by the weight ratio β/α — sweep the weights, sweep the phase from 0° (pure I) to 90° (pure Q).

Phasor View — Watching ϕ Move



Geometry = algebra:

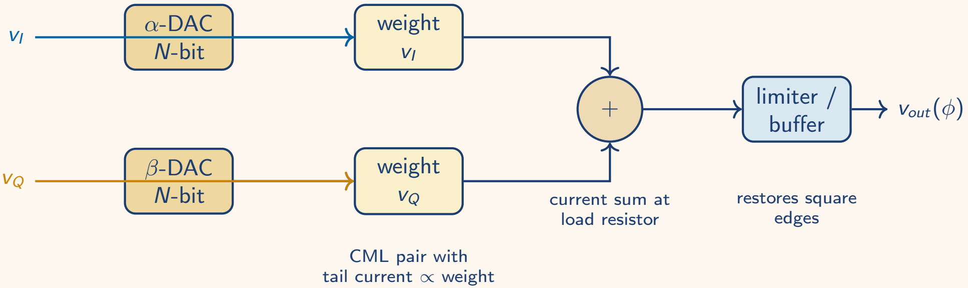
- α slides the tip along I
- β slides the tip along Q
- Tip's polar angle *is* ϕ

Two practical knobs:

Quadrant select choose which two of $\{I, Q, \overline{I}, \overline{Q}\}$ to sum $\rightarrow$ 360° coverage (coarse)

Weight ratio N-bit DAC sets $\beta/\alpha \rightarrow$ 2^N phase steps inside the quadrant (fine)

Practical Realization — A Visual Sketch



Two CML (current-mode logic) differential pairs share a load. Tail-current ratio $I_\alpha : I_\beta$ *is* the weight ratio $\alpha : \beta$. Sum is taken in the current domain — naturally linear, fast, and low-glitch.

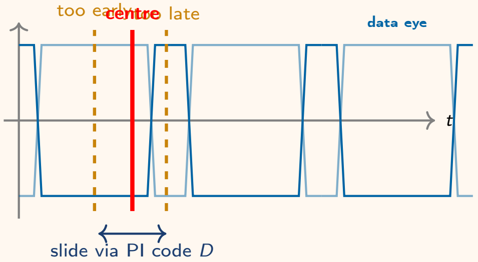
Applications

Application 1 — Wireline / SerDes CDR

Problem. Receiver gets serial data with no clock. Local reference clock is at the right *frequency* but unknown *phase*.

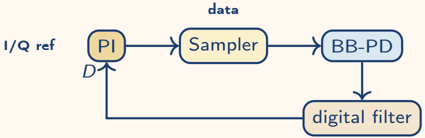
PI's job. Slide the sampling clock until it lands in the *centre of the data eye* — and keep it there as the link drifts.

$$\phi_{samp} = \frac{1}{2}\left(\phi_{rise} + \phi_{fall}\right)$$



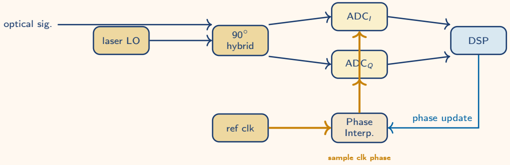
Why a PI?

Resolution of $360^\circ/2^N$ at multi-GHz rates with no PLL in the loop. A bang-bang phase detector dithers the PI code ± 1 LSB at lock — the entire CDR loop is digital after the sampler.



Application 2 — Optical Coherent Receivers

Setting. 100–800 Gb/s coherent links: optical front end delivers I and Q baseband (after mixing with a local laser) into ADCs running at 50 GS/s+.



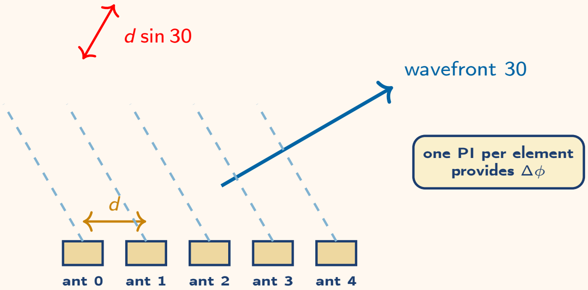
Where the PI lives.

- Generates the *ADC sampling clock phase* per lane
- Tracks slow phase drift between TX and RX symbol clocks with no analog VCO retuning
- Per-lane PI codes equalise inter-lane skew on parallel DSP fabric

Resolution matters

Clock jitter degrades SNR for 16/64-QAM. A 7-bit PI gives $\Delta\phi \approx 2.8^\circ \approx 80\,\text{fs}$ at 100 GHz.

Application 3 — Phased-Array Beam Steering



progressive delay $\Delta t = (d \sin 30)/c$

$\Rightarrow$ progressive phase $\Delta\phi = \frac{2\pi d}{\lambda} \sin 30$

Idea. Feed each antenna the *same signal* but with a progressive phase shift $\Delta\phi$ between neighbours. The array radiates a beam at angle

$$\boxed{\sin\theta = \frac{\lambda}{2\pi d}\,\Delta\phi.}$$

Change $\Delta\phi$ electronically $\to$ steer θ with no moving parts.

Why a PI per element?

- Continuous, monotonic phase
- Per-element skew calibration
- Same hardware across the band — only $\Delta\phi$ changes

Phase Resolution $\to$ Beam-Steering Resolution

Array equation (d spacing, λ wavelength). Differentiate to map small $\Delta\phi$ to small $\Delta\theta$:

$$\Delta\phi = \tfrac{2\pi d}{\lambda}\sin\theta \implies \boxed{\Delta\theta = \frac{\lambda}{2\pi d\cos\theta}\,\Delta\phi_{\mathrm{min}}.}$$

For an N-bit PI ($\Delta\phi_{\mathrm{min}} = 2\pi/2^N$) at the typical half-wavelength spacing $d = \lambda/2$: $\Delta\theta = (1/\cos\theta)\cdot 2^{-(N-1)}$ rad.

N	$\Delta\phi_{\mathrm{min}}$	$\Delta\theta_{@\,0^\circ}$	$\Delta\theta_{@\,60^\circ}$
4	22.5°	7.2°	14.3°
6	5.6°	1.8°	3.6°
8	1.4°	0.45°	0.90°

Two asymmetries

$1/\cos\theta$ blows up near endfire — same LSB buys coarser steps off-broadside. And PI *nonlinearity* (DNL/INL) sets the worst-case step, not the average.

Summary

- A PI is a weighted sum of two reference phases; the weight ratio sets $\phi = \arctan(\beta/\alpha)$, the frequency is unchanged.
- Geometry is just the phasor tip sliding inside an I/Q quadrant; coarse = quadrant select, fine = N-bit weight DAC.
- Realised as two CML pairs sharing a load — current-domain summation, no analog multiplier.
- **SerDes / optical:** sets the data sampling clock phase in fully digital CDR loops at GHz+ rates.
- **Phased arrays:** sets per-element progressive phase; $\Delta\theta = \frac{\lambda}{2\pi d \cos\theta}\Delta\phi$ ties phase resolution to angular resolution.
- Bit-count of the PI is the lever shared by every application — more bits buy finer eye-centering, lower DSP jitter, finer beam steps.

Questions?